Probabilistic Graphical Models (PGM)

Probability theory + Graph theory

Both theories are used in many areas of science & engineering to model real life phenomena and this particular class of machine learning models brings together the power of these 2 independent mathematical theories.

Most popular probabilistic Graphical Models are :-

* Bayesian Networks
* Markov Networks

Basic difference between the above two networks are :-

Bayesian n/w are directed acyclic graphs → eg:- decision tree
(DAGs)

Markov n/w are undirected graph, they could be cyclic.

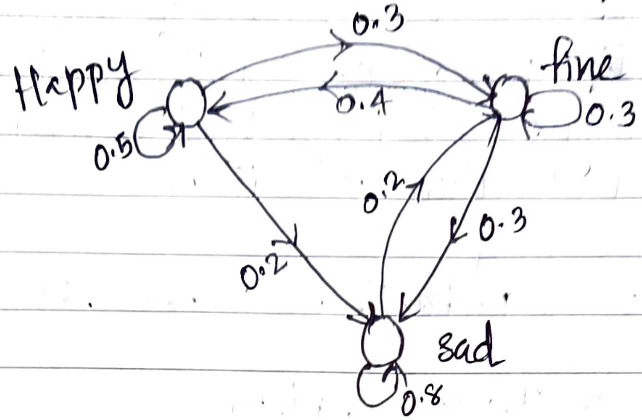
*→ DAGs are able to encode causal relationship and they are also easy to train, if you have given an amount of data DAGs can learn the underlying features lot more easily compared to undirected graphs.

*→ There are some problems which have cyclic dependencies and for that they have to use markov networks.

*→ PGMs are different from Markov chains.

Markov chain

Take a graph, 3 nodes - 3 emotional states that person could be in



Modelled the transition b/w these states using some pbblity. values.

Markov property usually means that your future state depends only on the present state and it doesn't matter which state you were in the past

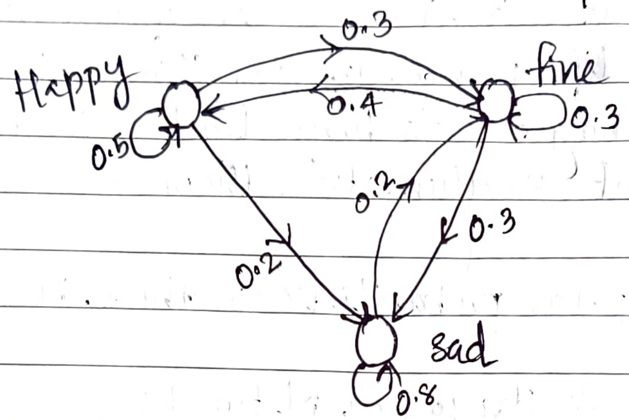
eg:- if you are fine there ~~you~~ will be ~~your~~ transition to various states with various probabilities but those probabilities donot depend in any way on what your previous state was and that is why this is called a Markov chain.

iii) PGM is very different from Markov chain, they look similar but the concept is very very different.

Reasons :- In Markov chain, the individual nodes are elements of a state space and edges denote the transition probabilities whereas in

Markov chain

Take a graph, 3 nodes - 3 emotional states that a person could be in



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ii, PGM is very different from Markov chain, they look similar but the concept is very very different.

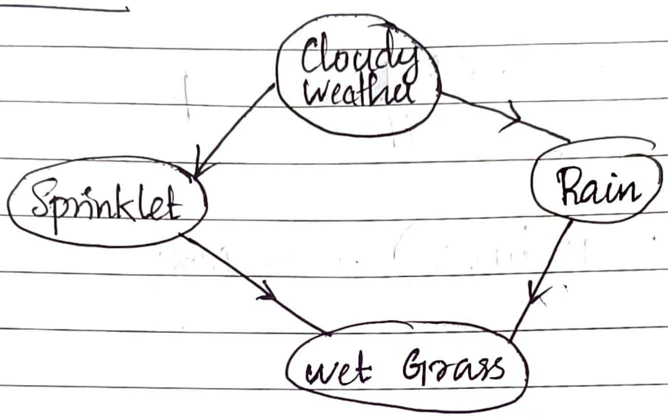
Reasons :- In Markov chain, the individual nodes are elements of a state space and edges denote the transition probabilities whereas in

PGIM the nodes are random variables and the edges denote conditional probabilities or causal relationships between different nodes.

rain
Switch on
sprinklet

Bayesian Network

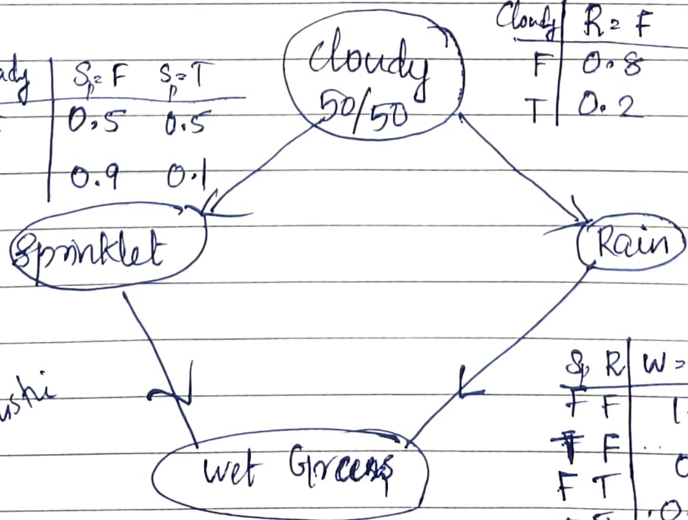
eg:-



Here edges doesnot represent the probabilities, it represent ~~the~~ only the causal relationship between two nodes.
cloudy and wet grass are not connected because there is no direct relationship b/w weather being cloudy and the grass being wet only if it rains or if the sprinklet is on.

Cloudy	R=F	R=T
F	0.8	0.2
T	0.2	0.8

cloudy	S_p=F	S_p=T
F	0.5	0.5
T	0.9	0.1



Pbblity table to make probabilistic inference

S	R	W=f	W=T
F	F	1.0	0.0
T	F	0.1	0.9
F	T	0.1	0.9
T	T	0.01	0.99

Estimate $P(S=T/W=T)$ - sprinkler being on given grass was found to be wet

Bayes theorem

$$= \frac{P(S=T) \,\&\, P(W=T)}{P(W=T)}$$

$$= \frac{\sum_{R,C \in T/F} P(S=T, W=T, R=T/F, C=T/F)}{P(W=T)}$$

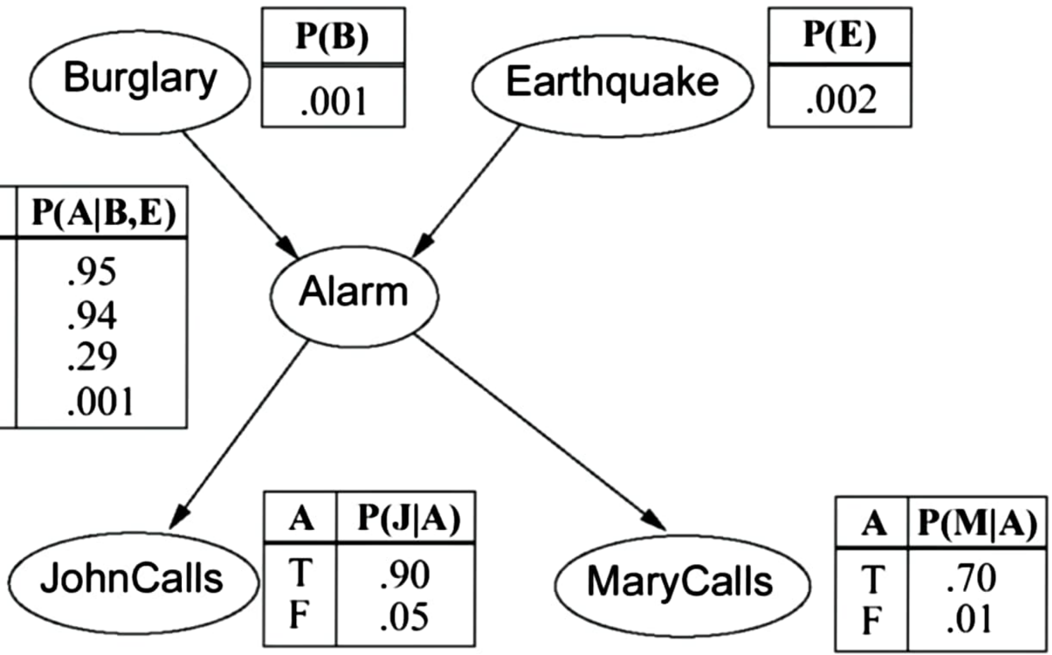
$$= \frac{0.2781}{0.6471} = \underline{\underline{0.430}}$$

$$P(R=T/W=T) = \underline{\underline{0.708}}$$

BAYESIAN BELIEF NETWORKS – EXAMPLE – 1

- You have a new burglar alarm installed at home.
- It is fairly reliable at detecting burglary, but also sometimes responds to minor earthquakes.
- You have two neighbors, John and Merry , who promised to call you at work when they hear the alarm.
- John always calls when he hears the alarm, but sometimes confuses telephone ringing with the alarm and calls too.
- Merry likes loud music and sometimes misses the alarm.
- Given the evidence of who has or has not called, we would like to estimate the probability of a burglary.

BAYESIAN BELIEF NETWORKS – EXAMPLE – 1



P(B)
.001

P(E)
.002

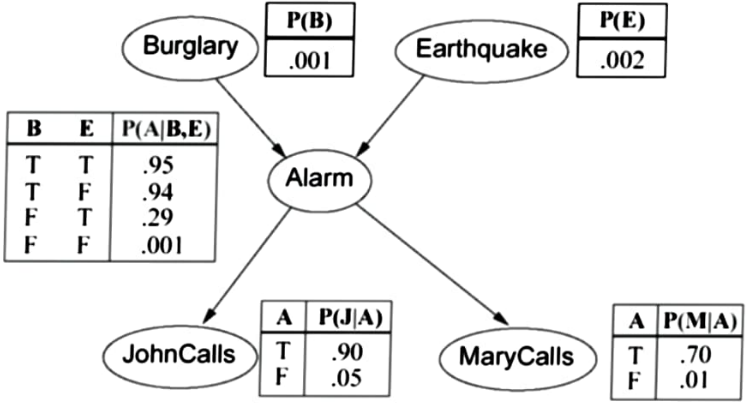
B	E	P(A|B,E)
T	T	.95
T	F	.94
F	T	.29
F	F	.001

A	P(J|A)
T	.90
F	.05

A	P(M|A)
T	.70
F	.01

BAYESIAN BELIEF NETWORKS – EXAMPLE – 1

1. What is the probability that the alarm has sounded but neither a burglary nor an earthquake has occurred, and both John and Merry call?



Solution:

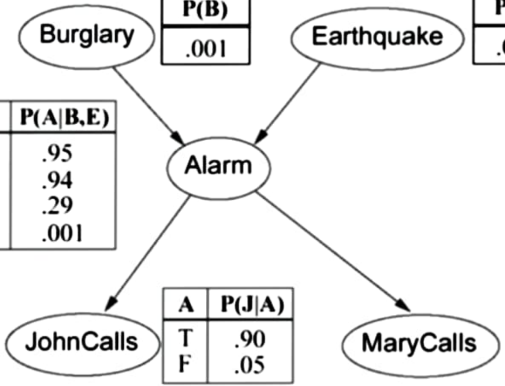
$$P(j \wedge m \wedge a \wedge \neg b \wedge \neg e) = P(j \mid a)\, P(m \mid a)\, P(a \mid \neg b, \neg e)\, P(\neg b)\, P(\neg e)$$

$$= 0.90 \times 0.70 \times 0.001 \times 0.999 \times 0.998$$

$$= 0.00062$$

BAYESIAN BELIEF NETWORKS – EXAMPLE – 1

2. What is the probability that John call?



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Solution:

$$P(j) = P(j \mid a)\,P(a) + P(j \mid \neg a)\,P(\neg a)$$

$$= P(j|a)\{P(a|b,e)*P(b,e)+P(a|\neg b,e)*P(\neg b,e)+P(a|b,\neg e)*P(b,\neg e)+P(a|\neg b,\neg e)*P(\neg b,\neg e)\}$$

$$+ P(j|\neg a)\{P(\neg a|b,e)*P(b,e)+P(\neg a|\neg b,e)*P(\neg b,e)+P(\neg a|b,\neg e)*P(b,\neg e)+P(\neg a|\neg b,\neg e)*P(\neg b,\neg e)\}$$

$$= 0.90 * 0.00252 + 0.05 * 0.9974 = 0.0521$$